

CIVITAS

Sustainable and smart mobility for all

2030



CIVITAS indicators

Bike-reserved paths connectivity index – Version 2 (TRA_BK_CN2)

DOMAIN

 Transport	 Environment	 Energy	 Society	 Economy
---	---	--	---	--

TOPIC

Bicycle

IMPACT

Cycling network connectivity

Improving the connectivity of the cycling network

TRA_BK

Category

Key indicator	Supplementary indicator	State indicator
----------------------	-------------------------	-----------------

CONTEXT AND RELEVANCE

Cycling is an environmentally and socially sustainable alternative to motorized private transport. Well-connected bike networks provide safe, direct routes that encourage more people to cycle, lowering carbon emissions associated to transportation. They support an efficient use of public space, as they reduce the need for extensive car parking infrastructure, and they foster public health by integrating physical activity and transportation. Bike networks can also bridge gaps in public transportation, improving access to transit hubs, workplaces and services. It is therefore desirable to develop well-connected urban bike networks to improve sustainability, accessibility, and health in urban environments.

This indicator provides a measure of the connectivity of the network of bike reserved paths. **It is a relevant indicator when the policy action is aimed at increasing the number of origin-destination pairs within a specific area of the city for which a bike route entirely on reserved paths exists. A successful action is reflected in a HIGHER value of the indicator.**

DESCRIPTION

This indicator is an index obtained as ratio between the **total length of bike-reserved paths in the experiment area** and the total length of roads within the experiment area. The indicator is dimensionless.

METHOD OF CALCULATION AND INPUTS

The indicator should be computed **exogenously**, by applying the method described and then coded in the supporting tool.

Method	
Calculation of the index based on the map of bike reserved paths and the map of roads	Significance: 0.50 
INPUTS The following information is needed to compute the indicator: a) A map of the roads on in the experiment area b) A map of the bike reserved paths in the experiment area where different bike path types are identified. Ultimately, types should allow to classify bike paths into three different categories: i. Bike lanes (bike path identified by lanes painted onto a street, not physically protected from lanes used by motorised vehicles) ii. Side path (side street bikeways with some kind of physical separation from motorised vehicles except at intersections) iii. Protected bikeway (infrastructure completely separated from roads used by motorised vehicles). The experiment would result in a modification of the map of bike reserved lanes in the experiment area either in terms of additional sections or in term of modification of lane type for some sections (e.g. from bike lane to side path).	
METHOD OF CALCULATION	

Commented [DF1]: Definitions and descriptions of bike paths categories to be checked/revised by an expert

Commented [LC2R1]: I would add a fourth type of infrastructure: cycle streets. These are roads where the carriageway is shared by motorized vehicles and bikes with no separation, but have low speed limits and narrow width. Cars are guests and should drive behind bikes at slow speeds. Ref: https://www.cyclinginwageningen.nl/news/bicyclestreet_Uvenweg I think this is relevant because here you imply that lower separation = worse conditions for cycling. However, cycle streets are supposed to be prime cycling infrastructure, as separation is not needed due to low car volumes on the link and, instead, it allows bikes to reclaim the whole space.

Commented [LC3R1]: On the three existing categories: I think clarity could be improved by using these terms -> bike lanes, protected side paths, segregated bikeways. There is no real consensus on the meaning of each term, but I would argue that protected means that there is some protection, while segregation means that the bikeway is completely decoupled from the road network.

The indicator should be computed exogenously according to the following steps:

- **Calculation of the total length of the road network within the experiment area.** This calculation can be obtained from the map of roads using a GIS application.
- **Calculation of the total length of the bike paths for each type within the experiment area.** This calculation can be obtained from the map bike paths using a GIS application.
- **Estimation of the index** by computing the ratio between a weighted sum of the lengths calculated in the second step (see equation below) and the length calculated in the first step.

EQUATIONS

The weighted sum of the lengths of the bike paths in the experiment area should be computed using the following equation:

$$WBkPLnght = 0.5 * BklnLnght + 0.75 * SdpthLnght + PrBkwLnght$$

Where:

BklnLnght = Length of bike lanes (see definition above)

SdpthLnght = Length of side paths (see definition above)

PrBkwLnght = Length of protected bikeways (see definition above)

The equation computing the index (last step of the method of calculation) is the following:

$$BkConnIndex = \frac{WBkPLnght}{TotRoadLnght}$$

Where:

WBkPLnght = weighted length of bike paths in the experiment area

TotRoadLnght = Total length of roads in the experiment area

Commented [DF4]: Text and/or equation could be modified after the check of bike paths categories above

ALTERNATIVE INDICATORS

This indicator assesses the connectivity of the bike-reserved network in an experiment area by measuring the share of length of bike paths out of total length of roads in the area, with different paths categories being weighted differently, to reflect higher or lower quality of bike infrastructure.

A simpler, albeit less significant version of this indicator is **TRA_BK_CN1**. It considers the total length of bike paths in the area, without weighting. This alternative indicator may be employed when data on the length of bike paths by category is unavailable. Both TRA_BK_CN1 and TRA_BK_CN2 can be used for local and whole city experiments.

TRA_BK_CN3 and **TRA_BK_CN4** measure the connectivity of bike-reserved networks accounting for the number of OD pairs that can be travelled cycling entirely on bike-reserved paths. They target local experiments and whole city experiments, respectively. These two indicators have higher significance since they reflect the actual usability of bike infrastructure for complete trips and capture network cohesion but require more complex input data.

	TRA_BK_CN1	TRA_BK_CN2	TRA_BK_CN3 (local) and TRA_BK_CN4 (whole city)
Complexity			
Significance			

It should also be noted that bike connectivity is one of the three components of **TRA_FC_AC1**. This indicator combines public transport connectivity, bike connectivity and road congestion to assess overall accessibility. The online tool automatically computes **TRA_FC_AC1** if the three sub-indicators have been calculated.